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Urban heat islands: a review of contributing factors, effects and data

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Abstract. Urban Heat Island (UHI) is one of the most serious difficulties that humans have faced in the twenty-first century resulted due to urbanization and industrialization. The main causes of UHI are the vast amounts of heat produced by urban structures, which solar energy should be used and re-radiated as well as anthropogenic heat sources. The two heat sources cause an urban area's temperature to rise in comparison to its surrounding areas, which refers to a phenomenon of the urban heat islands. Large cities and diverse economic activities face even more serious difficulties. This paper defined the concept and types of UHIs. This paper explained a comprehensive review of the phenomenon of UHI from its first appearance with regard to causing factors, its influences on people's lives and the urban environments, and widely used data for measuring and evaluating the intensity of UHIs. The finding of most studies mentioned that UHI values are different from city to city based on land-use changes. It is revealed that an increasing trend occurred mainly because expand in built-up areas and a reduction in green cover. Also, a number of studies showed that the UHI intensity was low during summer afternoons and high during winter nights in many places around the world.

1. Introduction

Recently, the phenomenon of urban heat islands (UHIs) has attracted many researchers, Luke Howard, a British scientist, first researched and documented the UHI phenomena in the 1810s when he observed that the city of London is warmer than the underdeveloped rural surrounds [1]. The phenomenon of an urban heat island (UHIs) is documented as rising air temperatures (AT) in cities compared to air temperatures in their countryside. Metropolitan heat islands are primarily created by the increasing use of land surfaces that trap heat in urban areas, for example, pavement, and have become an increasing concern as a result of rapid urban growth since the mid-twentieth century [2, 3]. Previous studies revealed that human comfort and health have been seriously influenced because of rising temperatures in many cities around the world. For example, Hong Kong, Bangkok, and Delhi witnessed an increase in mortality from 4.1 % to 5.8 % every 1 °C above a temperature threshold of roughly 29 °C in places [4]. The tendency of metropolitan regions to suffer greater temperatures in outdoors (up to 12 K higher) than their adjoining rural perimeter is referred to as the urban heat island (UHI) effect [5]. According to Organization (2016), half of the world's population lives in cities, and by 2030 urban regions are predicted to house 60% of the worldwide population, one in every three individuals residing in cities with populations of at least 500,000 people [6]. The UHI might be seen as a worldwide problem that endangers the functioning and habitability of cities and urban ecosystems [7]. Human comfort and health, air pollution in cities, energy management, and urban planning have all been impacted [8]. The UHI is affected by meteorological conditions as well as features of the surrounding environment such as materials, sky view factor, vegetation, and traffic [9, 10]. During heat waves, a lack of vegetation and

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surface wetness has been recognized as significant factors in the development of the UHI [11]. When metropolitan areas experience warmer temperatures than their rural surroundings, the UHI effect develops, the heat absorbed by constructed buildings and anthropogenic heat sources in a city are the major drivers of the UHI [12]. Recently, some studies showed that the UHI intensity was low during summer afternoons and high during winter nights [13].

2. General background of UHIs

2.1. Concept of UHIs

When cities replace the natural land cover with high concentrations of heat-absorbing pavements, buildings, and other surfaces, the UHI phenomenon develops. It is also a type of air pollution that contributes to global warming. When compared to the surrounding rural and suburban regions. UHI raises the temperature of metropolitan areas, such as city centres [14]. The UHI, a phenomenon in which temperatures in urban regions are greater than in non-urban areas, is a major anthropogenic change to the Earth's ecosystems [15]. UHI studies mainly depend on air temperatures, and land surface temperature (LST) retrieval from thermal infrared remote sensing data used in surface UHI studies. The difference between the AT within the city and the AT of its surrounds is referred to as UHI [15], traditional weather station automatic temperature measurements [16] or LST measurements provided by thermal infrared (TIR) remote sensing data are the two primary approaches to analyse the UHI phenomena. Mitigating the expected rise in winter temperatures, in addition to the frequency of exceptionally hot days and nights throughout the summer, are among the future difficulties [17].

2.2. Types of UHI

There are two types of UHI: atmospheric urban heat islands (AUHI) and surface urban heat islands (SUHI). The basic characteristics of the two kinds of heat islands are summarized in Table 1.

Table 1. Surface and atmospheric urban heat islands: Basic characteristics of UHIs [18].

Feature	Atmospheric UHI	Surface UHI
Temporal variation	May be small or non-existent during the day High intensty at night and in the winter	Day and night High intensty during the day and in the summer
Intensity of UHI	Less variation - Day (-1 to 3°C) - Night (7 to 12°C)	More spatial and temporal variation - Day (10 to 15°C) - Night (5 to 10°C)
Methods	Direct measurement: Fixed weather stations Mobile traver	Indirect measurement Remote sensing
Description	- Isotherm map - Temperature graph	- Thermal image

It is defined by warmer air in metropolitan regions in comparison to the colder air in surrounding rural areas. Experts frequently split heat islands into two categories [16, 20]:

- The canopy layer of the urban heat island (CUHI), extends upwards from the ground level to the ground mean height of buildings.
- The boundary layer urban heat island (BUHI) is a layer of air up to 2000 meters height above the canopy layer.

The canopy layer urban heat islands are the most prevalent and are frequently mentioned in the discussion of UHI. According to Basics (2011), in the late morning and throughout the day, AUHI is generally weak, whereas after sunset, due to the delayed release of heat from urban infrastructure the effect becomes more noticeable. The timing of this peak, on the other hand, is determined by the properties of urban and rural surfaces, as well as the season and weather conditions [18]. Figure 1 shows the types of UHI and the main components of the urban atmosphere.

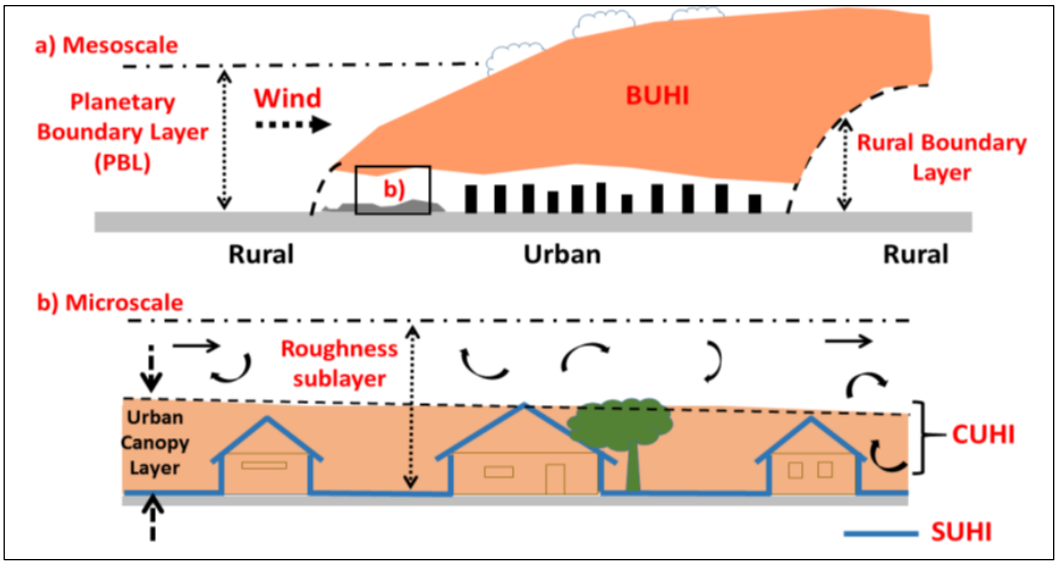


Figure 1. Types of UHI and the main components of the urban atmosphere [19].

SUHIs are defined as the variances between a core of a city and its surrounding areas within a certain radius based on LSTs [22, 23, and 24]. It is a significant manmade change of the surface of the Earth that has the potential to impact the local thermal climate by altering surface energy flow balances [22]. SUHI is primarily driven by variations in daytime radiative surface heating between urban and rural locations [23]. On a hot, bright summer day, the sun may heat dry, and exposed urban surfaces like roofs and pavement to temperatures (27 to 50 °C) higher than the air [24], while shaded or damp surfaces are often in more rural settings that remain near to AT. This can be found at any time of day or night, but they are most prevalent during the day when the sun shines. Daytime surface temperatures differ by (10 to 15°C) on average between developed and rural areas; whereas, nighttime surface temperatures differ by (10 to 15°C) [18]. Because of fluctuations in the intensity of the sun, as well as weather and the ground covers, the size of SUHIs varies with the seasons. SUHIs are often the greatest in the summer as a result of this variance [25].

3. Methods and data of UHIs

3.1. Air temperature

UHIs can be detected in the air below roofs (known as the CUHI). Heat islands in the canopy layer are mostly caused by direct site characteristics [26]. Weather balloons, aircraft equipment, and different remote sensing techniques are among the ways that may be used to observe the boundary layer [22,23]. Because the required tools are simple and easy to get, heat islands in the canopy layer have traditionally been the easiest to measure. Furthermore, canopy layer UHIs are thought to have the greatest impact on people since they present where most human activities occur. The AT discrepancies between the average of the AT measured in rural fixed masts (AT Rural) and the average of the AT recorded in urban fixed masts (AT Urban) were used to calculate the UHI impact [27], as shown in Equation (1).

$$UHI = AT_{Urban} - AT_{Rural} \quad (1)$$

Observations from fixed weather stations or transverse surveys undertaken in a few cities have helped. Understanding the three-dimensional impacts of urban shape on intra-city near-surface air temperature and wind flow provides crucial hints for mitigation measures focusing on urban planning and design [4]. Station observations, climate reanalysis datasets, and remote sensing are three methods for obtaining data. The most direct and accurate technique is station observation, which is always used as the ground truth to corroborate the AT calculated from other procedures. Long-time records and high observation frequency are two advantages of station observation. The longest AT records at certain stations can be over a century old [28].

3.2. Surface temperature

Although remote sensing-derived land surface temperatures (LSTs) are not equal to the temperature of the air above the earth, they are closely linked [26,27], when obtained using remote sensing data, however, SUHIs are what UHIs are called [29, 16]. SUHIs are defined as the differential in LSTs between a city or core city and its neighbouring regions inside the city at a certain range [17,31,32]. It computes using equation (2).

$$SUHI = LST_{Urban} - LST_{Rural} \quad (2)$$

Nonetheless, fast urbanization is most common in the countryside around cities, especially in big or megacities in developing countries. Furthermore, differences in LSTs between city centres and their rural areas were used in several earlier investigations. Rural areas were also divided into zones within a specified radius of a city; however, the distance is a factor. In prior investigations, the levels were too huge to monitor cities of various sizes. LST stands for distorted shortwave external reflectivity and is an important surface component. In recent years, the quality of LST data received from satellites has substantially increased. Then, several types of research on UHIs have been undertaken using LST remote sensing data from satellites, and these studies are referred to as SUHIs [3, 39, 40]. In recent decades, SUHI studies have received a lot of attention from researchers. In India, remote sensing data were used to study the diurnal, seasonal, and internal behaviour of SUHIs in 150 major cities around different climatic zones. This study employed the Moderate Resolution Imaging Spector-radiometer (MODIS) MODIS-Aqua and Terra (MYD11A2 and MOD11A2) satellites' 8-day global LST and emissivity product at spatial resolution with 1 km. According to the results of the correlation study and regression model, the SUHI phenomenon differs between Indian cities. Various variables in different climatic zones drive SUHI's diurnal and seasonal behaviour. The vegetation and evaporate-transpiration (ET) govern the latent heat flow over the surface and are shown to be primarily responsible for the daytime SUHI fluctuation in both seasons, with summer being more so than winter. Built-up intensity (BI) on the other hand, controls variance in night-time SUHI, especially in the summer [22]. In another study, retrieving and determining SUHI intensity values for 1449 Chin cities, the mean AT was partially positively correlated with night-time SUHII in 40% of cases using monthly LST data at a 1 km resolution and enhanced vegetation index data at 250 m resolution from MODIS/Aqua from December 2009 to November 2010. Central northern China had the highest accuracies, whereas the lowest was in western China during both daytime and night-time [28]. Multispectral satellite images with high spatial resolution were acquired over the Athens metropolitan region in Greece, and LST data was obtained from ASTER thermal data. The SUHI impact is noticeable on thermal maps during both daytime and night-time. In Athens metropolitan regions, for example, it is more obvious during the daytime [40]. Study in 2014, the spatial-temporal fluctuations of SUHIs across three metropolises in East Asia (Seoul, Tokyo, and Beijing) were examined using hourly LST data acquired from Korea's first geostationary multi-purpose satellite and COMS (Communication, Ocean, and Meteorological Satellite). Furthermore, a comparison of diurnal fluctuations between UHIs and SUHIs was investigated, as well as the influence of vegetation on the SUHI intensity. Accordingly, the intensity of SUHIs was higher during the day than at night, regardless of the cities studied [23].

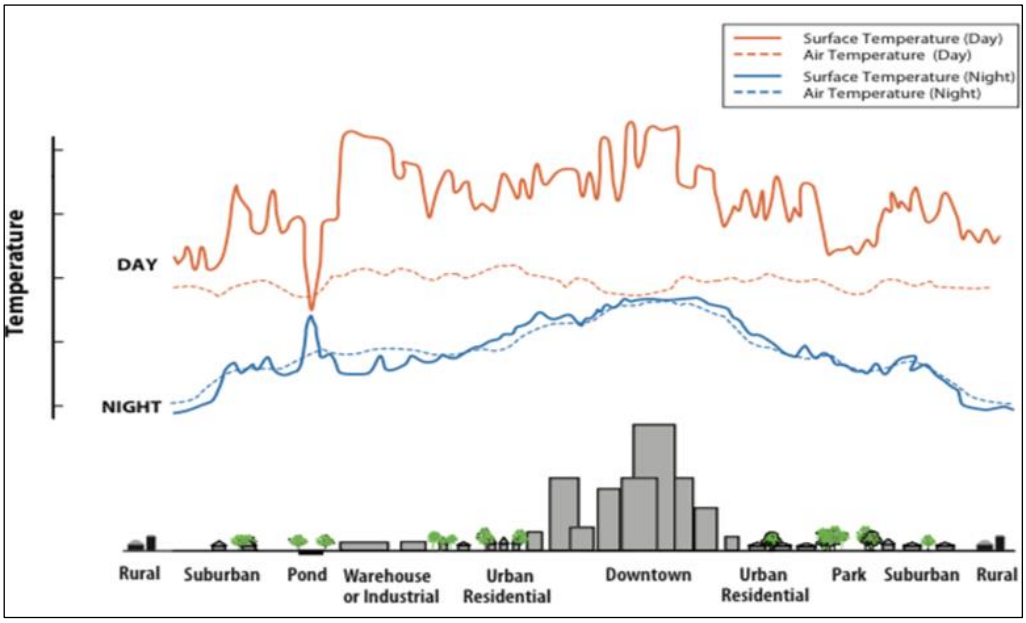


Figure 2. Atmospheric and surface temperature variation. (Variable land use zones have different surface and atmospheric temperatures. During the day, surface temperatures vary more than air temperatures, but at night, they are pretty comparable) [18].

4. Results and discussion

4.1. Contributing factors of UHIs

It is generally recognized that a variety of variables contribute to the rise in urban outdoor temperatures, as shown below:

4.1.1. Land use/ cover changes

Changes in land usage also influence the features of the land surface. Concrete and asphalt, for example, have considerably different thermal characteristics (thermal inertia/heat capacity) and surface radiative characteristics (albedo and emissivity) than materials usually utilized in rural regions [10]. Heat islands from asphalt and concrete are used to replace like asphalt and concrete are used to replace vegetation and natural surfaces on roadways, buildings, and other impervious surfaces. Urban materials change an urban surface's energy balance by absorbing instead of reflecting incoming solar energy, which raises surface temperatures as well as the overall ambient temperature. The energy balance is further influenced by the lack of vegetation in metropolitan areas, as natural cooling of the surface through evapotranspiration is reduced [40]. Differential heating occurs between impermeable surfaces, such as concrete and other heat-absorbing materials, and urban green areas and the surrounding naturally vegetated landscapes as a result of urbanization [23, 24, 3, and 25]. Urban geometry also has an impact on wind flow, energy absorption and emission due to the size of buildings and the distance between them in the city [18]. Because buildings function as wind barriers, the presence of built-up regions lowers wind speed by up to 60%. Reduced wind speed reduces heat convection from surfaces to the air, resulting in increased heat storage [41]. In addition, metropolitan areas tend to gather more net radiation than rural areas. Urban structures affect the levels of net radiation, as the walls of buildings absorb a large amount of solar radiation instead of allowing it to escape into the atmosphere, which leads to an increase in heat storage [26]. Ben-Dor and Saaroni (1997) found that LST variations between urban and rural locations may be as high as 5 °C and that these variances are mostly due to Urbanization and industrialization [42].

4.1.2. Anthropogenic activities

Anthropogenic activities are generated by human activity and can originate from a variety of places including industrial processes, buildings, automobiles, and people themselves all energy consumption. It can create from a multitude of places and is calculated by adding up all of the energy consumed to heat the house, appliances, transportation, and industrial activities are all run by electricity. The heat produced by anthropogenic activity and infrastructure varies depending on urban activity and infrastructure, with more energy-intensive buildings and transportation creating more heat [18]. The population parameter can refer to anthropogenic heat, as more people in a place means more energy consumption [43].

4.1.3. Other factors

Weather and location are two other elements that have a significant influence on UHI development and over which the community has limited control. Clear skies and calm breezes increase the amount of solar energy reaching the ground while reducing heat convection. Furthermore, geographical location influences climate and terrain. The existence of big watersheds, for example, decreases the above-air temperature, and the presence of mountains, if existent, may obstruct the winds [18]. The study of the SUHI phenomena in the Italian metropolis centres is strategically important since these geographical areas are home to roughly 30% of the Italian population. Furthermore, the Mediterranean region is regarded among the most popular significant "hotspots" of climatic change in future estimates, exacerbating the problem [47]. The population might have a direct and indirect influence on heat generation, as more people imply more metabolisms and an indirect effect as the number of structures increases. The number of cars, factories, and other facilities will almost certainly rise as a result of the growing population and urban expansion. There are additional factors to consider, such as building design, sky view, and properties of building material which have a role in generating heat contents; this is also explained by [44]. A recent study conducted (by Hamoodi, 2021) investigated how lands cover change and SUHIs were affected by armed and political conflicts [45]. Diurnal and seasonal variations of LST and SUHI were also investigated by several researchers around the world. For example, LST maps were prepared using Landsat images to analyse variation in surface temperature between daytime and nighttime for summer and winter in the city of Baghdad [48]. In 2020, a study was conducted in Seoul, North Korea, where the main focus of this study was to contribute to the examination of the determinants of air temperature from the diurnal and seasonal perspectives. Data were collected from January 2015 to December 2016 from the Korean Meteorological Administration [49]. Also, another study calculated the difference in urban land surface, diurnal and seasonal in the city of Lagos, which is located in the Gulf of Guinea, where LST data were collected from MODIS. The study found that the contribution of urban LSTs was lower at night than during the day, possibly due to differences in the quantity of solar energy received during the day and night [51]. In 2021, a study was conducted in Bangladesh focused on the diurnal and seasonal fluctuations of SUHI. LST data were collected from MODIS from 2003 to 2019, the magnitude of the SUHI fluctuated substantially between day and night, while the magnitude was higher during the day than at night. The diurnal and seasonal amplitudes of surface temperature narrowed and broadened, particularly during the winter season. In this context, the temporal composition of the daytime urban surface cooling island SUCI, and the effect of nighttime SUHI in Erbil were assessed. The data were collected from LST and vegetation index (NDVI) from MODIS from December 2003 to 2004. The findings showed that highly built-up regions had lower LST functioning as a SUCI during the day, in summer, autumn, and winter than non-urbanized areas outside the city. While at night Erbil had a large SUHI effect and suffered increased LST [50]. Several studies examined the SUHI in two or more cities. A study carried out by Jasim Mohammed Ali (2017) stated that the SUHI's intensity in London was greater than in Baghdad despite Baghdad having greater diurnal, daytime, and night-time LSTs than London and Birmingham [22]. Another study mapped the regional patterns of LST over Mumbai using Landsat 5 TM data in 2010. The data showed that the city's centre and southern areas had the greatest temperatures throughout the day, while the periphery had lower temperatures [46].

4.2. Impacts of UHIs

UHI have import effects in several aspects, as explained in the following sections.

4.2.1. Human health and comfort

Hot or cold weather can have a severe influence on human health, exacerbating respiratory or cardiovascular diseases and increasing the chance of death. The UHI effect puts urban populations at a higher risk of heat-related illnesses. Human health impacts due to UHI might cause human mortality and disease, for example, the temperature of the land surfaces may reach 60 ºC during the summer. Another illustration from the United States of America (USA), the Centres for Controlling Disease and Prevention claims that UHI between 1979 and 2003 was responsible for almost 8,000 fatalities in the USA [43]. This has induced the researchers to model the human heat balance. The variation between the surface and the air causes massive colonization of the air; this can lead to nasal bleeding due to the lack of air zones and high pressure on earth's surfaces. Researchers in South Asia studied changes in human comfort levels caused by heat islands, using a variety the indexes of thermal comfort such as to analyze outside circumstances, using the Physiologically Equivalent Temperature, Temperature Humidity Index, and Relative Strain Index. The PET parameter was computed by Emmanuel and Johansson values to assess the ambient factors that impact thermal comfort [17, 31]. Emmanuel and Johansson calculated Physiologically Equivalent Temperature values to assess the ambient factors that impact thermal comfort. Thermal discomfort was found to be considerable, especially on clear days, because of a mix of high temperatures, low wind speeds, and strong solar radiation. Wide roadways with low-rise structures with no trees to provide shade had the poorest conditions. The most pleasant circumstances were discovered on small streets with tall buildings, particularly when shade trees were available, and along the shore, where the sea breeze was beneficial [4].

4.2.2. Energy consumption

Because of the rise in temperatures in cities, the energy demand for cooling grows in tandem with rising energy consumption. According to Oke, every 0.6°C raises in AT raises the power consumption for cooling by 1.5–2.0% in the United States of America (USA). In addition, 5-10% of power output is utilized to compensate for cities' continuously rising temperatures [25]. As a result, air pollution rises in tandem with the rise in energy generation. For example, in the United States of America (USA), fossil fuels are mostly used to generate power. Sulfur Dioxide (SO2), Nitrogen oxide (NOx), and other pollutants emitted by most power plants contribute to greenhouse gas emissions and hence to global warming and climate change [43]. In terms of water quality, SUHI damages it mostly through thermal pollution, resulting in harmful aquatic ecosystems. According to James, Initial rainfall temperatures may be raised from around (21°C) to cover (35°C) by using (38°C) pavements [56]. Ali pointed out that heated stormwater runoff causes a change in water temperature in aquatic environments, which may be harmful or even fatal to aquatic animals [43].

4.2.3. Other effects

SUHIs are mostly because by thermal pollution, which damages water quality. The temperature of the water has an impact on aquatic life in all aspects, particularly various aquatic species' metabolism and reproduction. Warm storm water runoff can cause rapid temperature variations in aquatic ecosystems, which can be particularly stressful. When the water temperature fluctuates more than (1- 2°C) in 24 hours, brook trout, for example, incur thermal shock and thermal stress [18]. According to the Chinese Ministry of Environmental Protection's "Ambient air quality standards", certain monitoring stations on the ground surpassed 280 g/m3 (140 ppb), exceeding 40% of the permissible value of 1-hour average O3 concentration of 200 g/m3 (100 ppb). In addition, photochemical fog is primarily composed of O3. Due to its strong oxidizing properties, tropospheric ozone pollution has been shown to have a negative impact on human health and agricultural growth. Increased air temperatures enhance the rate of ground-level ozone generation, which is created in the presence of sunshine, NOX and volatile organic compounds react. In addition to increased air emissions, Ozone emissions at ground level will be greater

in sunny and hotter weather if all other variables remain constant, such as precursor emissions or wind speed and direction [18].

5. Conclusions

This paper was conducted to summarize and review the basic concepts of the phenomena of the UHI to determine the contributing factors, effects and common data used in measuring and evaluating the phenomenon. It is considered one of the most important problems facing urban areas, as they have been defined as a phenomenon in which temperatures in urban regions are greater than in non-urban areas [25]. Therefore, this study presented first the factors that contribute to developing this phenomenon, which included urban development, human activities and some other factors. As the changes in land use also influence the features of the land surface. Concrete and asphalt, for example, have considerably different thermal characteristics (thermal inertia/heat capacity) and surface radiative characteristics (albedo and emissivity) than surfaces usually covered in rural regions. Heat islands from replacements like asphalt and concrete are used to replace vegetation and natural surfaces on roadways, buildings, and others needed to accommodate growing populations. Building materials change an urban surface's energy balance by absorbing instead of reflecting incoming solar energy, which raises surface temperatures as well as the overall ambient temperature. This increase in temperature influences human health and comfort, and energy consumption. A variety of datasets have been used in studying the different types of the phenomenon of UHI mainly including weather stations for measuring atmospheric UHI and remotely sensed data (i.e., thermal data) for measuring surface UHI.

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